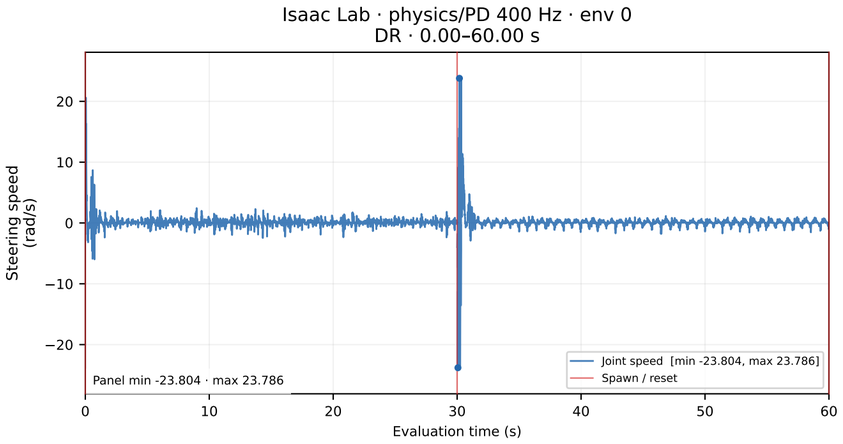
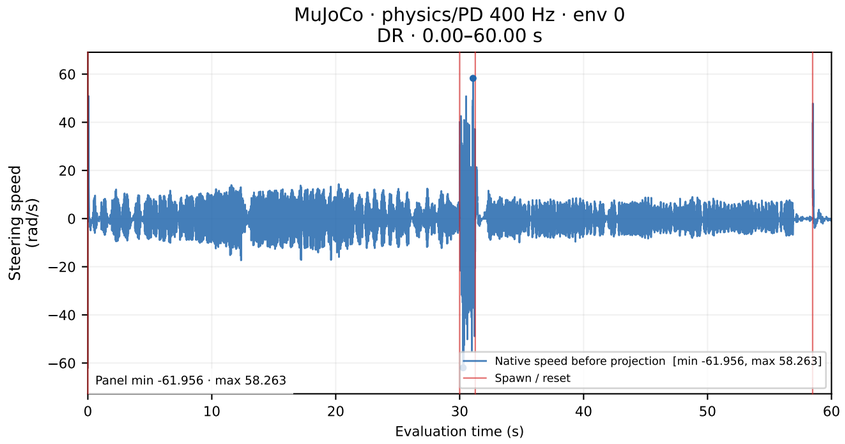
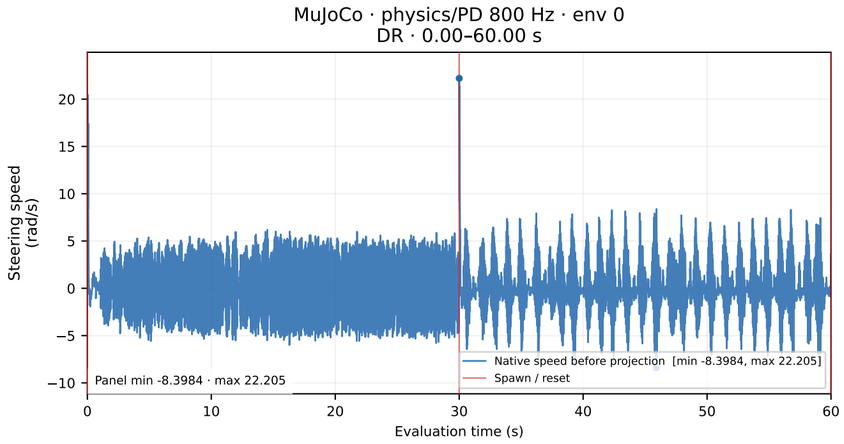
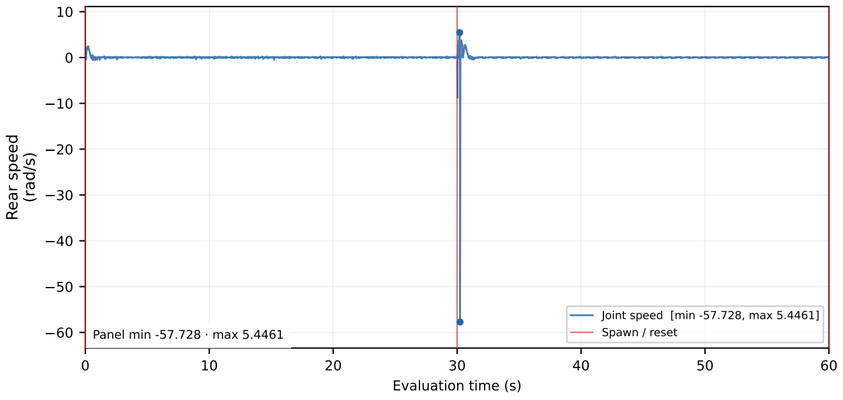
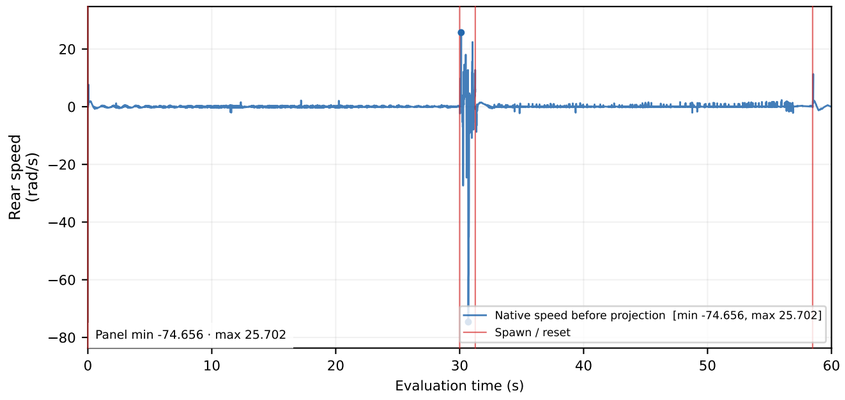
model_2500 · Physics-substep joint speeds · env 0 · full 60 s

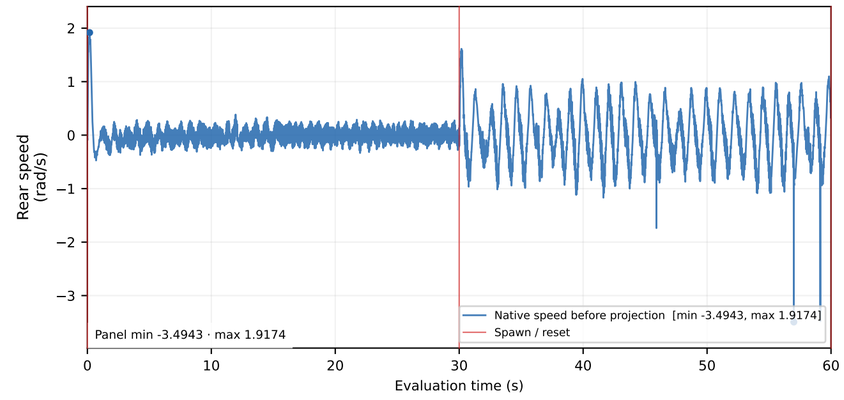












50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
MuJoCo physics velocities are BEFORE its hard post-step speed projection; observations use the projected state.
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 0; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04518, 0.804024, 0.959366, 0.816123]
  CoM XYZ mm [4.87718, -3.711, 0.621036]
  mu_s [0.741039, 0.681711, 0.592046, 0.532645]
  mu_d [0.614405, 0.611756, 0.534947, 0.532645]
  Kp [steer,rear] [10.6517, 0]; Kd [0.209184, 0]
  spawn roll 0.0198978 rad; rate 0.00125125 rad/s
  gyro bias XYZ rad/s [-0.000195088, -0.000150685, -2.64955e-05]
  gravity bias YZ [0.00115356, 0.00192213]
E2 [30.00, 60.00) s:
  mass x [1.07174, 1.18127, 1.11706, 0.881481]
  CoM XYZ mm [-0.792429, -4.35209, 3.72319]
  mu_s [0.576972, 0.659445, 0.531594, 0.547743]
  mu_d [0.576972, 0.547619, 0.531594, 0.547743]
  Kp [steer,rear] [9.66427, 0]; Kd [0.207538, 0]
  spawn roll -0.0845213 rad; rate 0.112094 rad/s
  gyro bias XYZ rad/s [-2.89794e-05, -3.58017e-05, 0.000157862]
  gravity bias YZ [-0.0029177, -0.00080667]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
E2 [30.00, 31.26) s:
  mass x [1.02693, 1.08918, 0.965679, 0.927341]
  CoM XYZ mm [1.65679, 1.84224, -2.29571]
  mu_s [0.541939, 0.732187, 0.611239, 0.700521]
  mu_d [0.534359, 0.631664, 0.611239, 0.641329]
  Kp [steer,rear] [10.5106, 0]; Kd [0.202627, 0]
  spawn roll 0.059515 rad; rate -0.0147133 rad/s
  gyro bias XYZ rad/s [-4.78947e-06, 5.43307e-05, 0.000350425]
  gravity bias YZ [-0.000315206, -0.00188852]
E3 [31.26, 58.48) s:
  mass x [0.831615, 1.11129, 0.86509, 1.17322]
  CoM XYZ mm [4.90593, -3.7759, 1.83006]
  mu_s [0.669571, 0.546287, 0.69316, 0.640867]
  mu_d [0.669571, 0.546287, 0.69316, 0.556841]
  Kp [steer,rear] [9.35915, 0]; Kd [0.186665, 0]
  spawn roll -0.0194645 rad; rate -0.0189324 rad/s
  gyro bias XYZ rad/s [9.16669e-05, -2.25439e-05, 8.03725e-05]
  gravity bias YZ [0.00145273, 0.000686634]
E4 [58.48, 60.00) s:
  mass x [1.14858, 0.965632, 0.91743, 0.960669]
  CoM XYZ mm [-4.05657, 3.38234, 4.15215]
  mu_s [0.723429, 0.71008, 0.69316, 0.546287]
  mu_d [0.723429, 0.59371, 0.69316, 0.546287]
  Kp [steer,rear] [10.2283, 0]; Kd [0.182062, 0]
  spawn roll 0.085862 rad; rate 0.244715 rad/s
  gyro bias XYZ rad/s [-8.09711e-05, 0.000283962, 0.000268881]
  gravity bias YZ [-0.000732425, -0.000166156]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 0; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.05389, 1.02143, 1.02368, 0.92158]
  CoM XYZ mm [-4.69182, -0.632826, -2.85415]
  mu_s [0.666375, 0.640016, 0.669571, 0.71155]
  mu_d [0.519145, 0.513141, 0.669571, 0.71155]
  Kp [steer,rear] [9.46788, 0]; Kd [0.191255, 0]
  spawn roll 0.0628041 rad; rate -0.166514 rad/s
  gyro bias XYZ rad/s [0.000138097, -8.54505e-05, 3.17079e-05]
  gravity bias YZ [0.00125118, -0.000618693]
E2 [30.00, 60.00) s:
  mass x [1.02693, 1.08918, 0.965679, 0.927341]
  CoM XYZ mm [1.65679, 1.84224, -2.29571]
  mu_s [0.541939, 0.732187, 0.611239, 0.700521]
  mu_d [0.534359, 0.631664, 0.611239, 0.641329]
  Kp [steer,rear] [10.5106, 0]; Kd [0.202627, 0]
  spawn roll 0.059515 rad; rate -0.0147133 rad/s
  gyro bias XYZ rad/s [-4.78947e-06, 5.43307e-05, 0.000350425]
  gravity bias YZ [-0.000315206, -0.00188852]
Six significant digits shown; full precision: sampled_dr_values.json.
```